**SES x Codera Hackathon information sheet**

## Task

Please create a monthly exchange rate forecast using EconData (and any other data source you wish to use) to produce 1-month ahead USDZAR forecasts every Friday (out of sample) and calculate forecast errors based on RMSEs for your model and a AR(1) model for the period 1 January 2021 to 31 December 2025.

Details:

- To ensure comparability, please use the USDZAR provided by SARB (from EconData (South African Reserve Bank -> Rates -> Market Rates -> EXCX135)
- Training/Estimation data can go back as far as you would like, or your data allows.
- Describe your model, the data it uses, present summary statistics such as observation count for your estimation and forecast period.
- Plot a chart of the RMSEs from your model and the AR1, noting the numbers on the chart.
- Demonstrate that your forecasts are based on an out of sample approach.
- Also provide plots of the series you use in your model.
- You may use AI, but you need to explain how your code and the model works.

- You may use more than one model, but we will judge the most accurate model.
- Code will need to shared for marking purposes.

**Rules**

• Eligibility: Teams of 2-4 (Stellenbosch University students only). Register via the poster link.

• Tools: Any software is allowed. AI is permitted, but you must be able to fully explain your code.

• Integrity: Zero tolerance for copying other teams.

• Judging: Submissions will be evaluated by a panel of 4 judges, subject to change

• Attendance: Info session (July 24, 3-5 PM) is highly recommended. At least 1 team member must attend the prize-giving (July 28, 6-9 PM) to claim a prize.

• Prizes (per team): 1st: R4000 | 2nd: R2000 | 3rd: R1000.

**Presentation**

- Each team will present their model to the panel of judges, including a powerpoint presentation
- after the presentation, the judges will have the opportunity to ask questions
- The deadline for submission is Monday 11:59pm . No late applications will be considered.
- The submission form must be completed and sent to 28399323@sun.ac.za